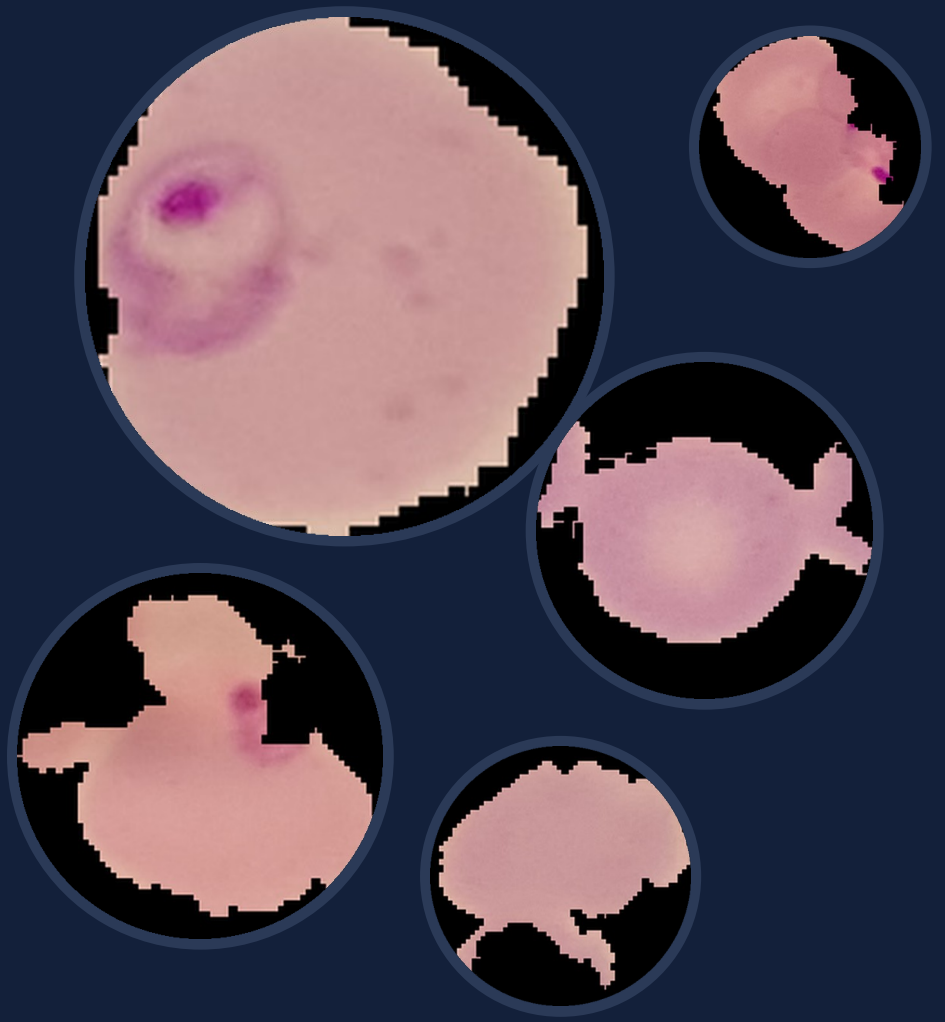
PHASE 3 · FINAL PRESENTATION

Does 98 % survive unseen patients?

Reproducing an EfficientNet-B0 ensemble for malaria cell detection from its authors' own code, then testing it on patients it has never seen



J Amathziah
230139

C Murali Madhav
230115

Ravi Yadav
230131

Pugazhendhi J
230066

Department of Computer Science · Rishihood University, Sonipat · github.com/amathziah/malaria-detection

THE ANSWER, UP FRONT

Reproduced within 1 pp, then tested on unseen patients

97.62 %

ensemble accuracy with the paper's own recipe, 10 folds, paper protocol (paper: 98.29 %; every metric within 1 pp)

98.45 %

the same recipe on 40 patients no model ever saw (patient-level split, clean evaluation)

−0.26 pp

change in MCC from YUV stain normalisation on unseen patients

Yes, largely: the ensemble scores 98.45 % on 40 held-out patients. Measured on identical cells, patient leakage is real but small (0.85 pp of MCC). YUV stain normalisation does not help: its effect stays within ±1 pp. H1 is not supported; H2 is rejected.

THE PROBLEM

Microscopy is the gold standard, and it doesn't scale

Slow

A thin smear holds hundreds of cells; a technician checks them one by one.

Expert-dependent

It needs a trained microscopist, scarce exactly where malaria is common.

Subjective

Counts and calls vary between observers and between laboratories.

Automatable

Single-cell images are a well-posed vision task: parasitized or uninfected.

Goal: classify single red-blood-cell images as parasitized or uninfected, accurately on new patients, and light enough to run where it is needed.

THE DATA

27,558 cells from 200 patients

27,558

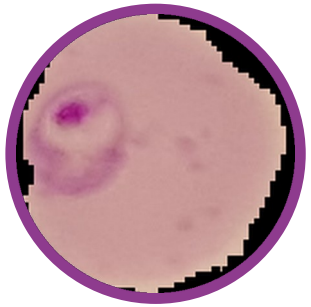
single-cell images, 50 % parasitized

200

patients: 150 infected, 50 healthy (Chittagong Medical College Hospital)

75 %

of uninfected cells come from infected patients' smears



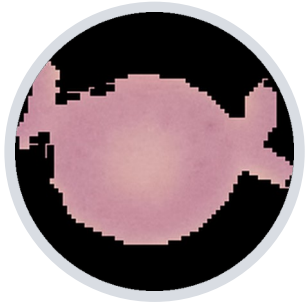
Parasitized



Parasitized



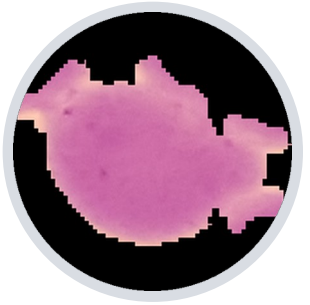
Parasitized



Uninfected



Uninfected



Uninfected

The patient is in the file name

`C33P1thinF_IMG_2015…_cell_179.png`

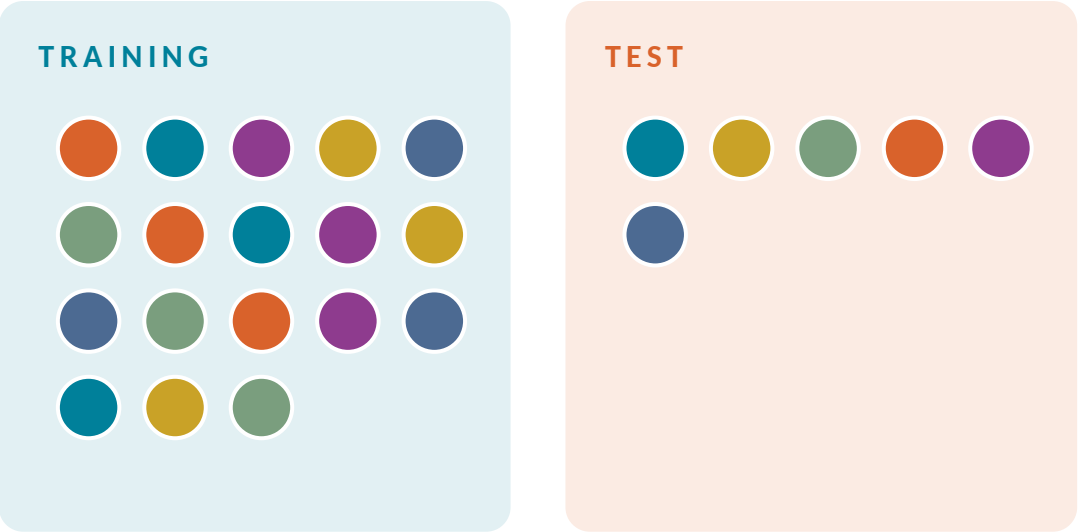
C33 = patient 33. We checked this key against NIH's own patient-to-cell mapping: all 27,558 cells map to 200 patients.

THE TRAP

A random split puts the same patient on both sides

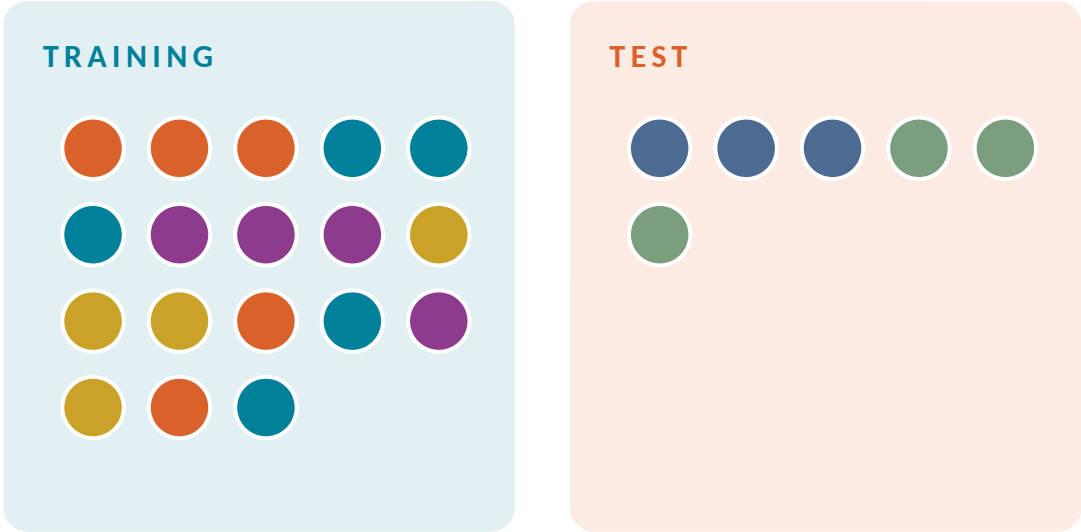
Image split (the paper)

cells shuffled at random: every patient on both sides



Patient split (ours)

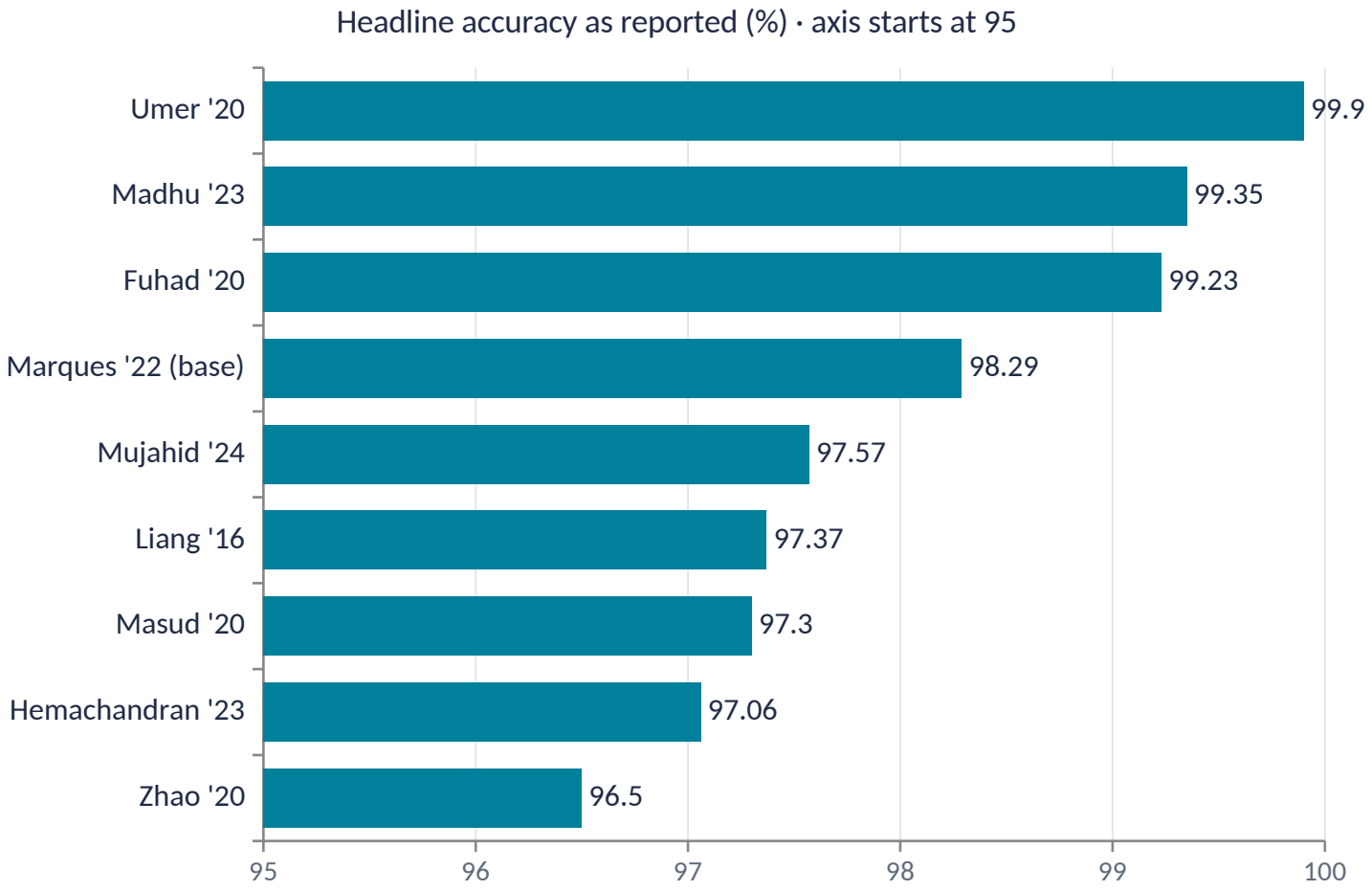
whole patients go to one side only



Each colour is one patient. A patient's cells share stain, lighting and blood, so under an image split the model can recognise the patient instead of the parasite. Only 2 of the 11 studies we reviewed evaluate on unseen patients.

PHASE 1 · 11 STUDIES

High reported accuracy, rarely measured on new patients



Patient leakage

Most split images at random; only Rajaraman et al. ['18, '19] test on unseen patients.

Not comparable

5-fold, 10-fold and 80:20 protocols; original or cleaned labels.

Rarely reproducible

Marques et al. published their executed notebook, so we could rebuild it exactly.

PHASE 1 · REPRODUCIBILITY CHECKLIST

Strong on code and data, weak on the split and evaluation

Checklist item	Paper	What Marques et al. give	What we did
Data public	Yes	NIH Malaria cell images, 27,558 crops	Downloaded automatically; patient IDs checked against NIH's mapping
Code public	Yes	The authors' executed notebook (supplementary file)	Rebuilt from the code, not from the text
Model size	Yes	Table 2: 4,223,934 parameters	Matched exactly; an automated test checks it
Hardware and time	Yes	GTX 1050, 4 h 46 min per fold	RTX 4090, about 11 min per fold
Hyperparameters	Partly	The text differs from the code in 8 places	We follow the code (next slide)
Library versions	Partly	Only in the notebook (TensorFlow 2.3)	TensorFlow 2.15, the last Keras 2; same efficientnet 1.1.1
Metrics	Partly	Precision and recall swapped in the text	Recomputed from the printed reports
Data split	No	Random 80/20 and 10 folds, unseeded	Sorted and seeded; a fingerprint is stored with every fold
Evaluation protocol	No	Not stated; the code scored augmented images	We report clean and augmented evaluation
Patient-level test	No	Random image split only	Became our hypothesis H1
Trained models	No	Not released	All 40 of our models are public on GitHub

Every gap we could close, we closed: a seeded split, both evaluation protocols, and all trained models public.

PHASE 2 · BASE PAPER

The authors' code differs from their paper's text

Setting	Paper's text / our first guess	Authors' code (what we follow)
Weights	ImageNet	noisy-student, qubvel efficientnet 1.1.1
Head	GAP → dropout → 1 sigmoid	GAP → 128 → 64 → 32 (dropout 0.3) → 2 softmax
Loss	binary cross-entropy	categorical cross-entropy, label smoothing 0.1
Schedule	≤ 15 epochs, early stopping	33 epochs, LR halved on plateau, best checkpoint
Batch size	128	16 (1,240 steps per epoch)
Hold-out	10 %	20 % = 5,512 images, never trained on
Augmentation	flips, 90° rotations	heavy albumentations: noise, blur, distortions, CLAHE…
Evaluation	clean images	augmented images (one generator for all)

32 automated tests check the rebuild against the paper's printed outputs (e.g. its 4,223,934 parameters).

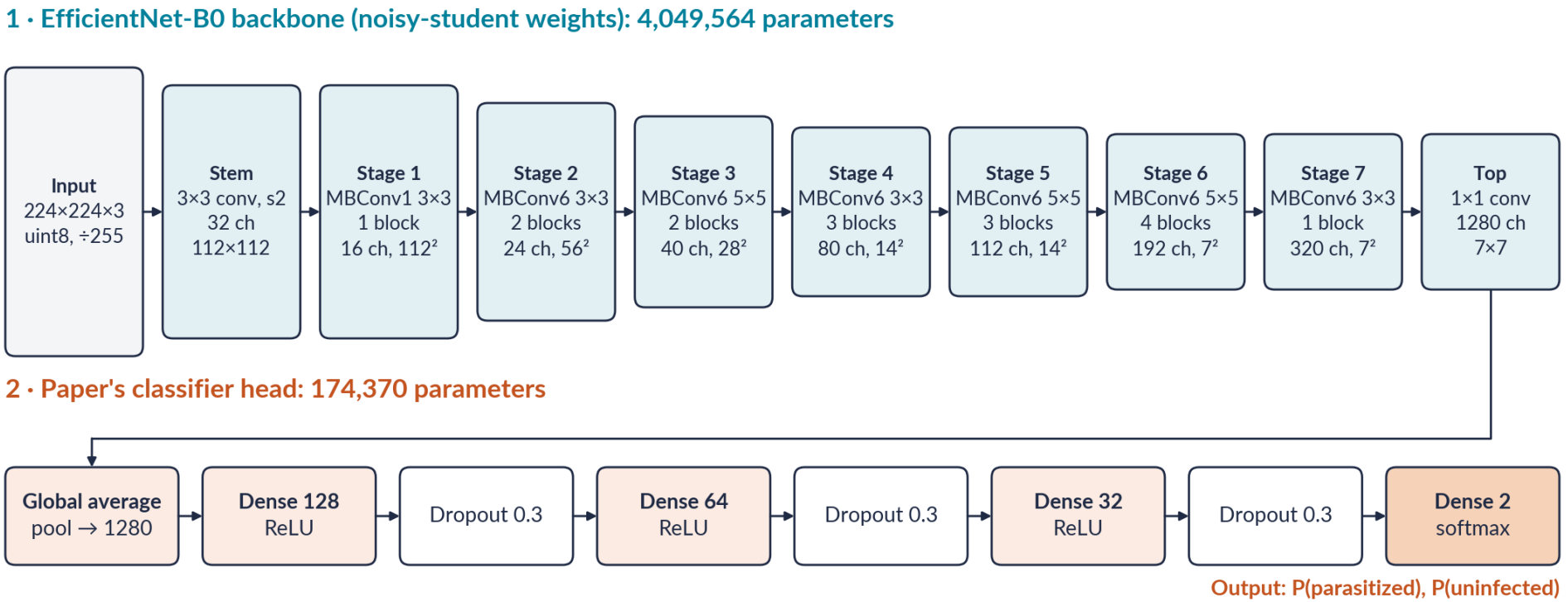
Two things the paper's numbers hide

Precision and recall are swapped in the text: the true ensemble recall is 97.74 %, not 98.82 %.

Every reported prediction was on an augmented image. We report both protocols; it changes our numbers by ≤ 0.23 pp.

THE MODEL

EfficientNet-B0 with the paper's head: 4.2 M parameters



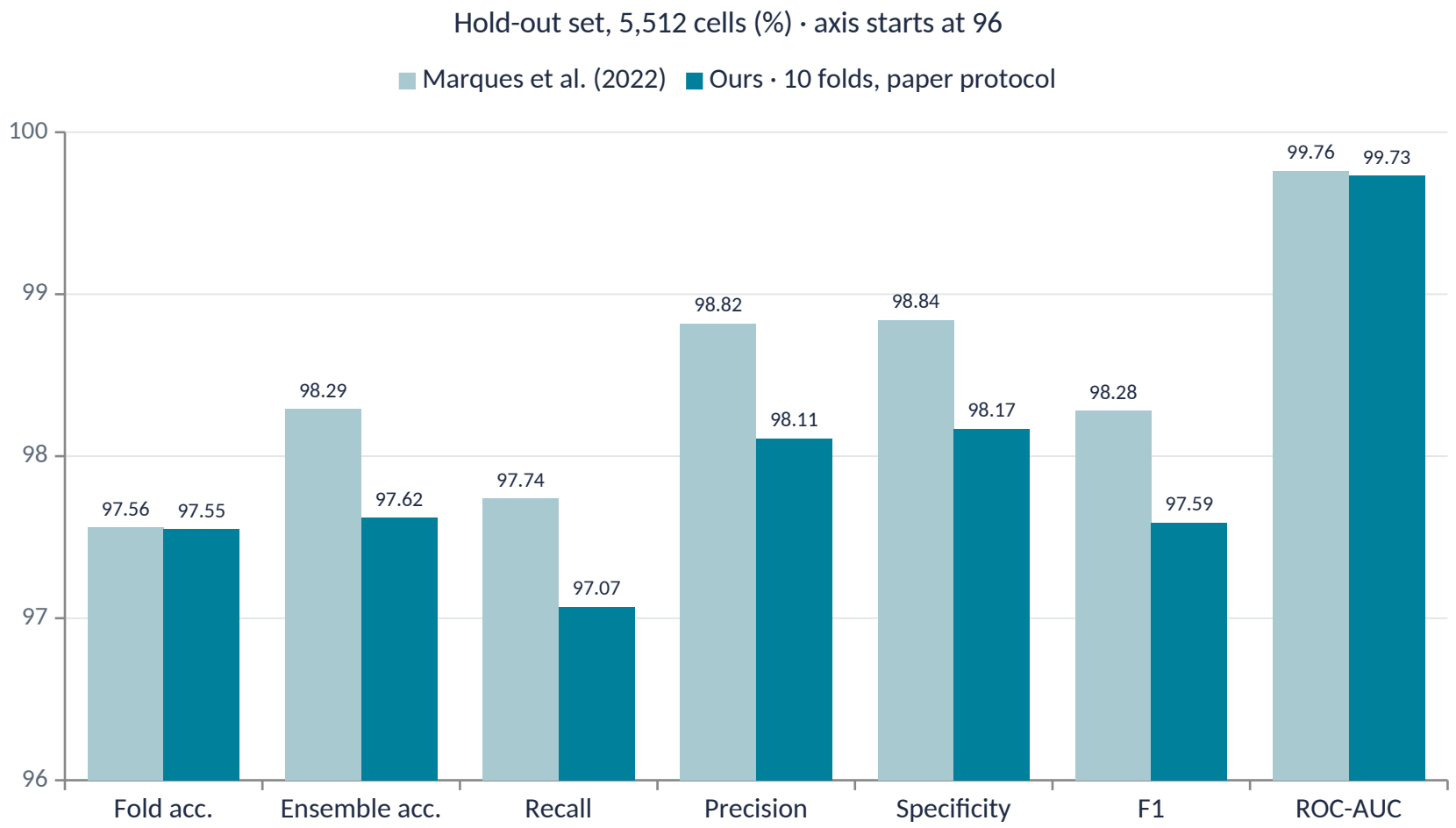
Box height shows the resolution (224 → 112 → 56 → 28 → 14 → 7 pixels) as the channels grow 3 → 32 → … → 320 → 1280.
16 MBConv blocks in 7 stages · 81 convolution layers (65 normal + 16 depthwise) · 49 batch-norm layers
230 Keras layers in the backbone · 4 dense layers in the head

Total: 4,223,934 parameters (4,181,918 trainable + 42,016 non-trainable batch-norm statistics), exactly the paper's Table 2.

224 × 224 cell → 16 MBConv blocks shrink the image to 7 × 7 while growing 3 colour channels into 1,280 feature maps → three dense layers → P(parasitized), P(uninfected). Ten such models, one per fold, are averaged.

PHASE 2 · RESULT

A faithful reproduction: every metric within 1 pp



−0.01 pp

fold accuracy, 97.55 vs 97.56 %

−0.67 pp

ensemble accuracy: 97.62 vs 98.29 % (≈ 3 standard errors)

Ensemble on 5,512 hold-out cells: 80 infected cells missed, 51 healthy cells flagged (paper: 62 and 32).

ENGINEERING

Ten GPUs in parallel: 11 min per fold, not 4 h 46 min

≈ 25×

faster per fold, same computation

≈ 11 min

per fold on an RTX 4090 (paper: 4 h 46 min on a GTX 1050)

10

RunPod GPUs at once, one fold each; all 40 trained models are public on GitHub

Decode once

27,558 images resized once into a shared uint8 cache

Parallel, seeded augmentation

worker processes, each batch seeded by (fold, epoch, batch)

Bytes to the GPU

uint8 batches, scaled by 1/255 inside the model

XLA-compiled step

fused kernels: 21 → 12.5 ms per training step

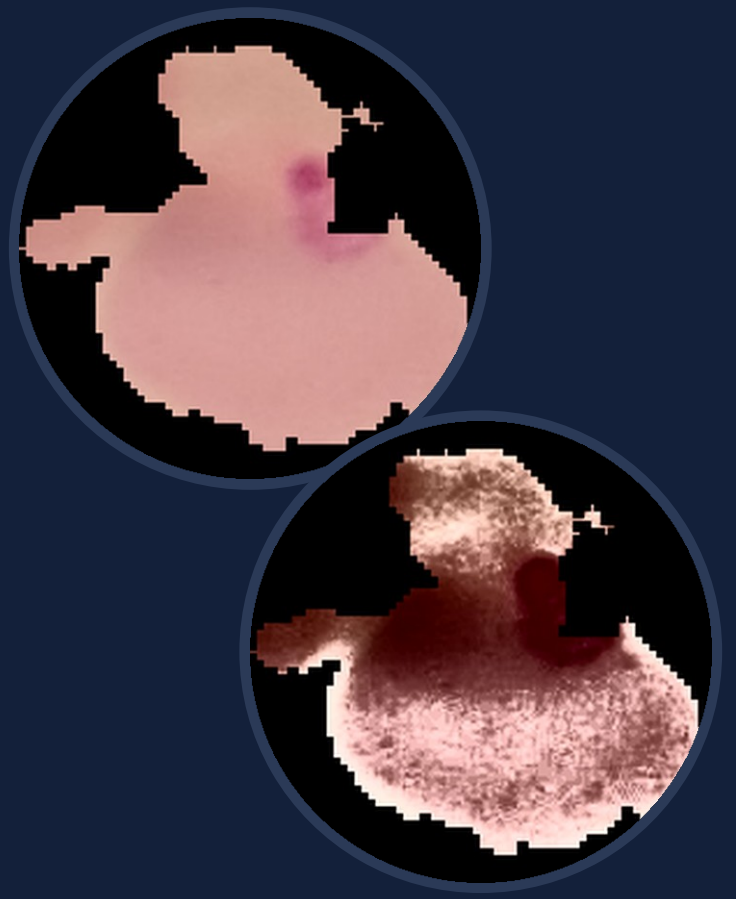
Same maths

float32 throughout; results identical for any worker count

PHASE 3

Two hypotheses, one change at a time

Registered in our Phase 2 deck, with the analysis plan committed before any new model was trained



WHAT WE TEST

H1: leakage inflates scores · H2: normalisation helps

H1 · VALIDATION

Split by patient, not by image

We expect Accuracy and MCC fall below the image-split baseline; fold-to-fold spread widens.

Because Cells of one patient share stain, lighting and blood: a random split lets the model recognise the patient.

H2 · PREPROCESSING

YUV + histogram equalisation (Umer et al.)

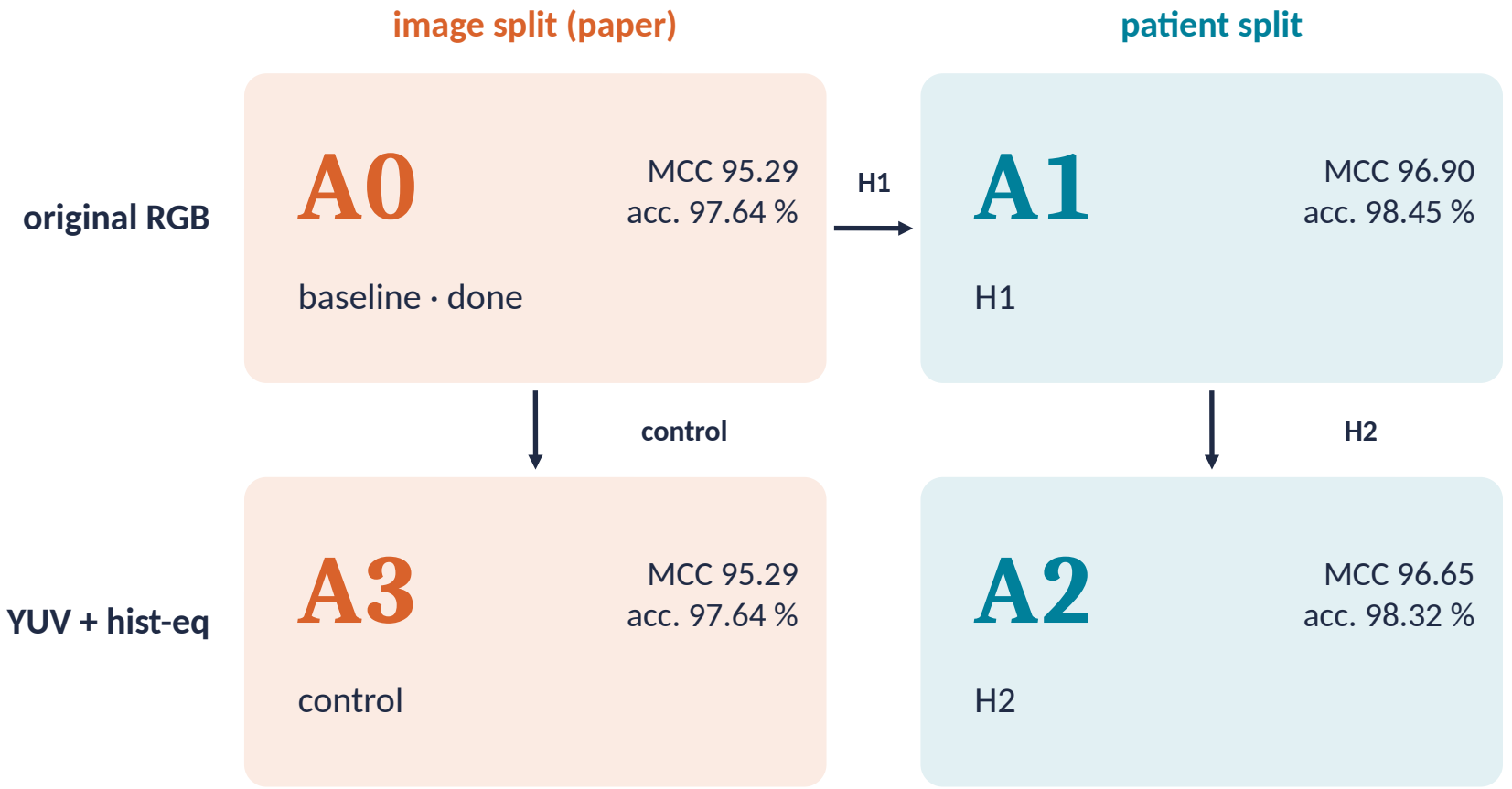
We expect Higher recall and MCC than H1 on unseen patients; little change on the random split.

Because Equalising brightness removes slide-specific cues, so the network must rely on the parasite.

Falsifiable: if H2's gain over H1 is smaller than the fold-to-fold SD, we reject H2. If patient and image splits score the same, we reject H1.

ABLATION DESIGN

A 2 × 2 design: each arrow changes one thing



Held fixed

model and weights
loss and optimiser
33 epochs, batch 16
augmentation
per-fold seeds
10-model ensemble
A1 = A2 split
A3 = A0 split

THE TWO SWITCHES

40 held-out patients and a stain-normalised input

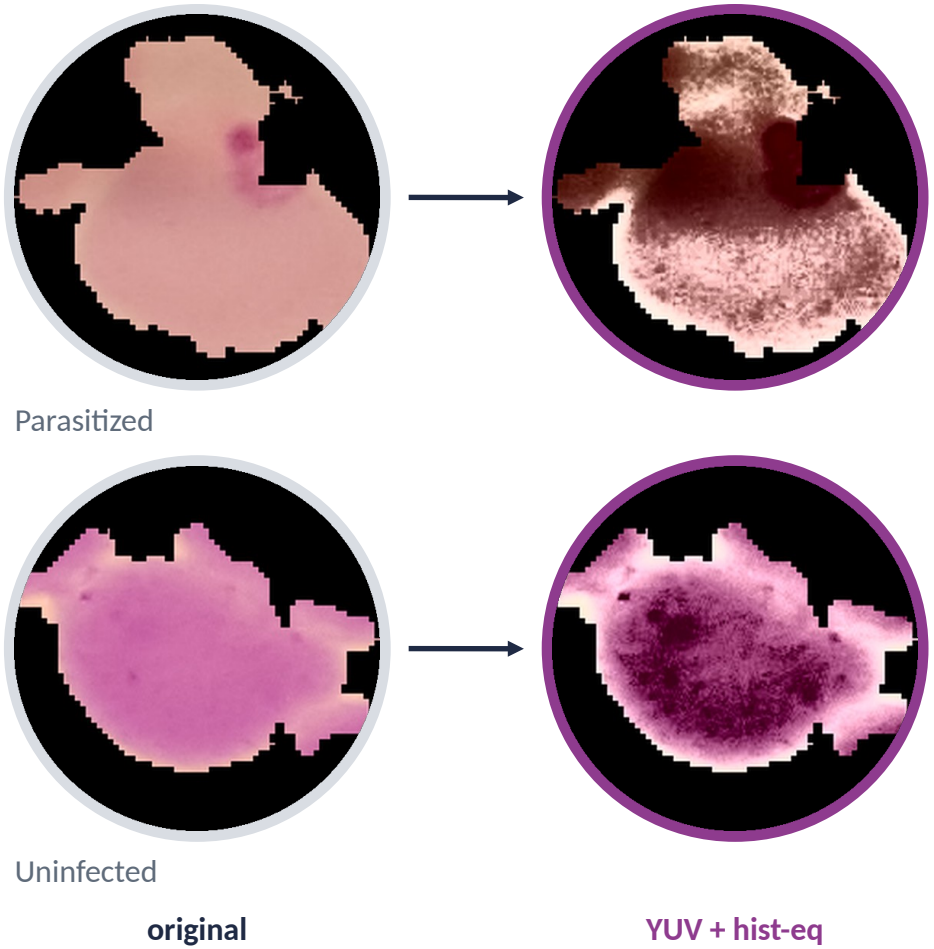
40

patients held out (28 infected): 5,550 cells, 50.7 % parasitized

160

patients in 10 folds of 15 to 17 patients, StratifiedGroupKFold

Split committed to the repository; every run checks its fingerprint.



How the YUV step works

1 RGB → YUV (OpenCV)

2 equalise the brightness (Y) histogram

3 YUV → RGB, colour unchanged

4 the paper's resize and augmentation

It stretches each cell's narrow brightness range to the full range, which also amplifies texture.

HOW WE DECIDE

Patients, not cells, are the unit of evidence

Patient bootstrap

Every interval resamples whole patients (10,000 times), because a patient's cells are not independent.

Paired where possible

A2 vs A1 and A3 vs A0 compare the same cells: corrected resampled t-test over folds (Nadeau & Bengio) and a paired bootstrap of the ensemble.

Holm correction

Three contrasts, so p-values are corrected for testing three times.

Fixed in advance

Primary metric MCC, clean evaluation, decision rules: committed to GitHub before the runs. Later analyses: a dated addendum, before running them.

RESULTS AT A GLANCE

Every metric, all four experiments

	A0 · image split	A1 · patient split	A2 · patient + YUV	A3 · image + YUV
Hold-out set	5,512 cells, 200 patients	5,550 cells, 40 patients	5,550 cells, 40 patients	5,512 cells, 200 patients
Ensemble accuracy	97.64 %	98.45 %	98.32 %	97.64 %
Recall (sensitivity)	**97.18 %**	**98.05 %**	**98.22 %**	**97.14 %**
Specificity	98.09 %	98.87 %	98.43 %	98.13 %
MCC × 100	95.29	96.90	96.65	95.29
ROC-AUC	**99.68 %**	**99.80 %**	**99.76 %**	**99.66 %**
Missed / false alarms	77 / 53	55 / 31	50 / 43	78 / 52
Fold MCC, mean ± SD	95.57 ± 0.60	93.92 ± 1.51	93.32 ± 1.66	95.20 ± 0.37

Ten-model ensembles, clean evaluation, each on its own hold-out set: A1 and A2 share the 40 unseen patients; A0 and A3 share the paper's split, whose test cells come from all 200 patients. 95 % patient-bootstrap intervals are in the paper (Table III). Under the paper's augmented protocol A0 scores 97.62 % (slide 10).

Unseen patients cost little: H1 is not supported

95.3 → 96.9

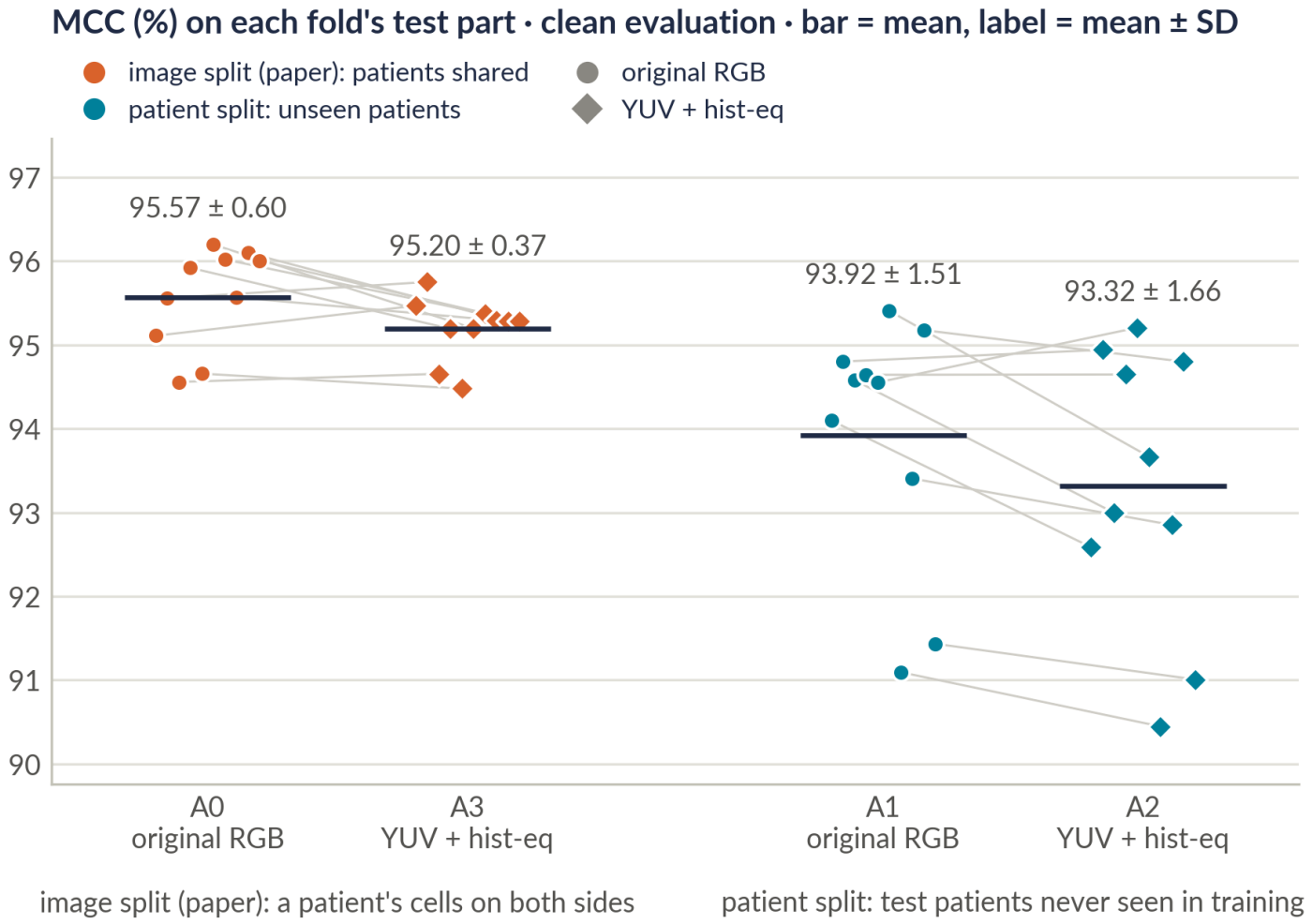
ensemble MCC × 100: image split (A0) → patient split (A1)

+1.62 pp

ensemble MCC on unseen patients, 95 % CI +0.01 pp to +3.12 pp; Holm p = 0.15

−1.65 pp

fold MCC over the 160 CV patients (95 % CI −3.28 pp to −0.02 pp; Holm p = 0.14): small and fragile



Scored on the same cells, leakage is real but small

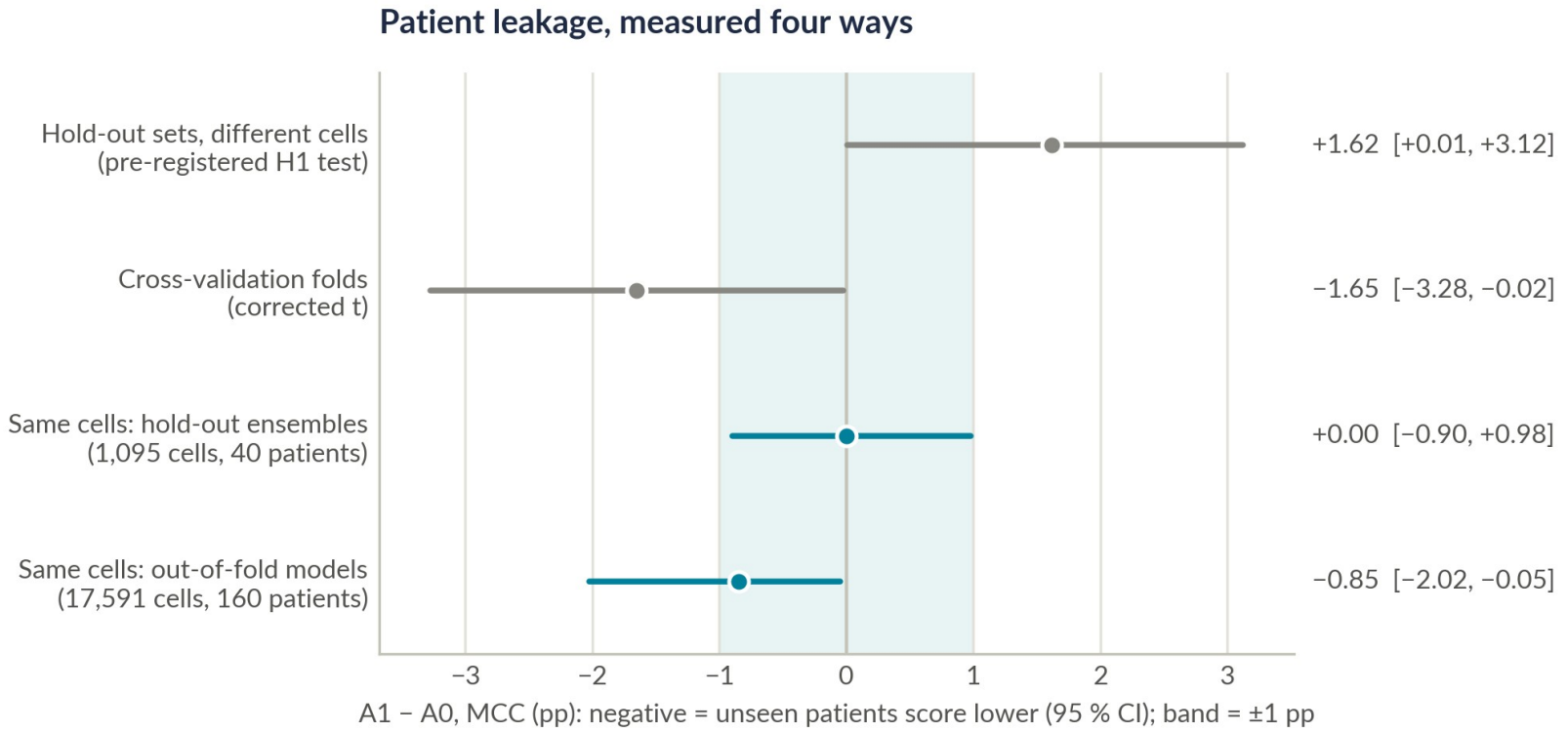
−0.85 pp

MCC on 17,591 cells of 160 patients, out-of-fold models (95 % CI −2.02 pp to −0.05 pp)

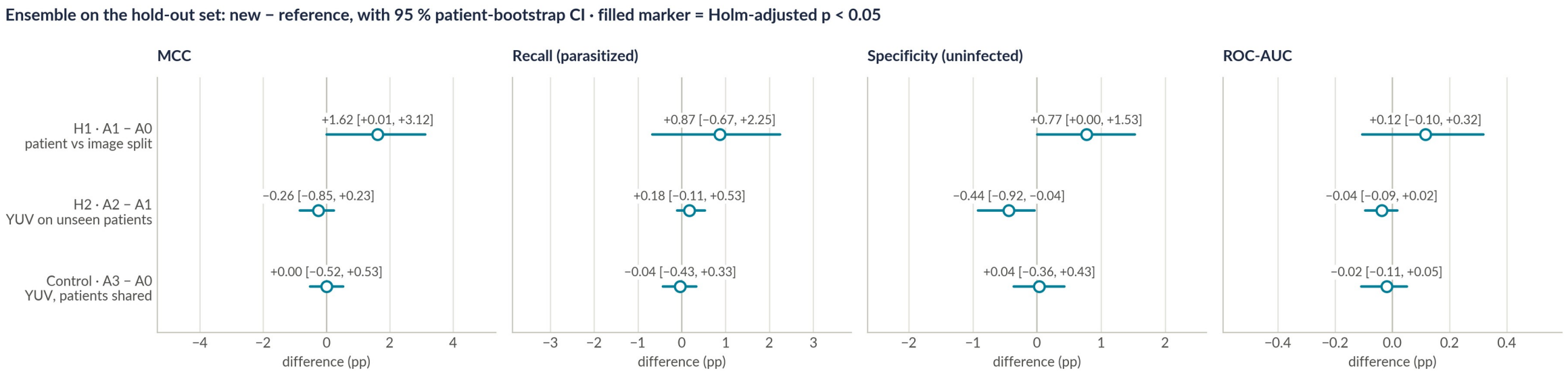
+0.00 pp

MCC on 1,095 cells of the 40 held-out patients, ensembles (95 % CI −0.90 pp to +0.98 pp)

Same cells, two models: one had seen other cells of the patient (A0), the other never saw the patient (A1).



YUV normalisation doesn't help on unseen patients: H2 is rejected



H2 · A2 vs A1 (same 40 unseen patients)

Mean fold MCC −0.60 pp vs A1's fold SD 1.51 pp (pre-registered rule: reject). Ensemble MCC −0.26 pp, 95 % CI −0.85 pp to +0.23 pp: equivalent to zero within ±1 pp (90 % CI −0.73 pp to +0.16 pp).

Control · A3 vs A0 (patients shared)

Mean fold MCC −0.37 pp vs A0's fold SD 0.60 pp. Ensemble MCC +0.00 pp, 95 % CI −0.52 pp to +0.53 pp: YUV changes little when patients are shared.

PREDICTIONS VS OUTCOMES

Both predictions were wrong, and we can say why

H1 · PATIENT LEAKAGE

Predicted a clear drop on unseen patients, and a wider fold-to-fold spread.

Observed the spread widened 2.5× as predicted, but the drop is small: 0.85 pp of MCC on identical cells, and none on the 40-patient hold-out set.

Likely why the decisive cue, purple-stained parasite chromatin, looks alike in every patient, and the paper's heavy colour, contrast and blur augmentation already hides each patient's stain.

H2 · YUV STAIN NORMALISATION

Predicted higher recall and MCC on unseen patients.

Observed MCC −0.26 pp, within ±1 pp of zero; recall +0.18 pp and specificity −0.44 pp, neither significant after correction.

Likely why the step equalises brightness only, not stain colour; the augmentation already varies brightness and contrast; and Umer et al.'s large gain was measured against a model that had not trained.

What we learned: measure a gain against a strong baseline, and measure leakage on identical cells. Next test: colour augmentation on another lab's smears [16].

FAILURE ANALYSIS

Errors cluster in a few patients

7 of 40

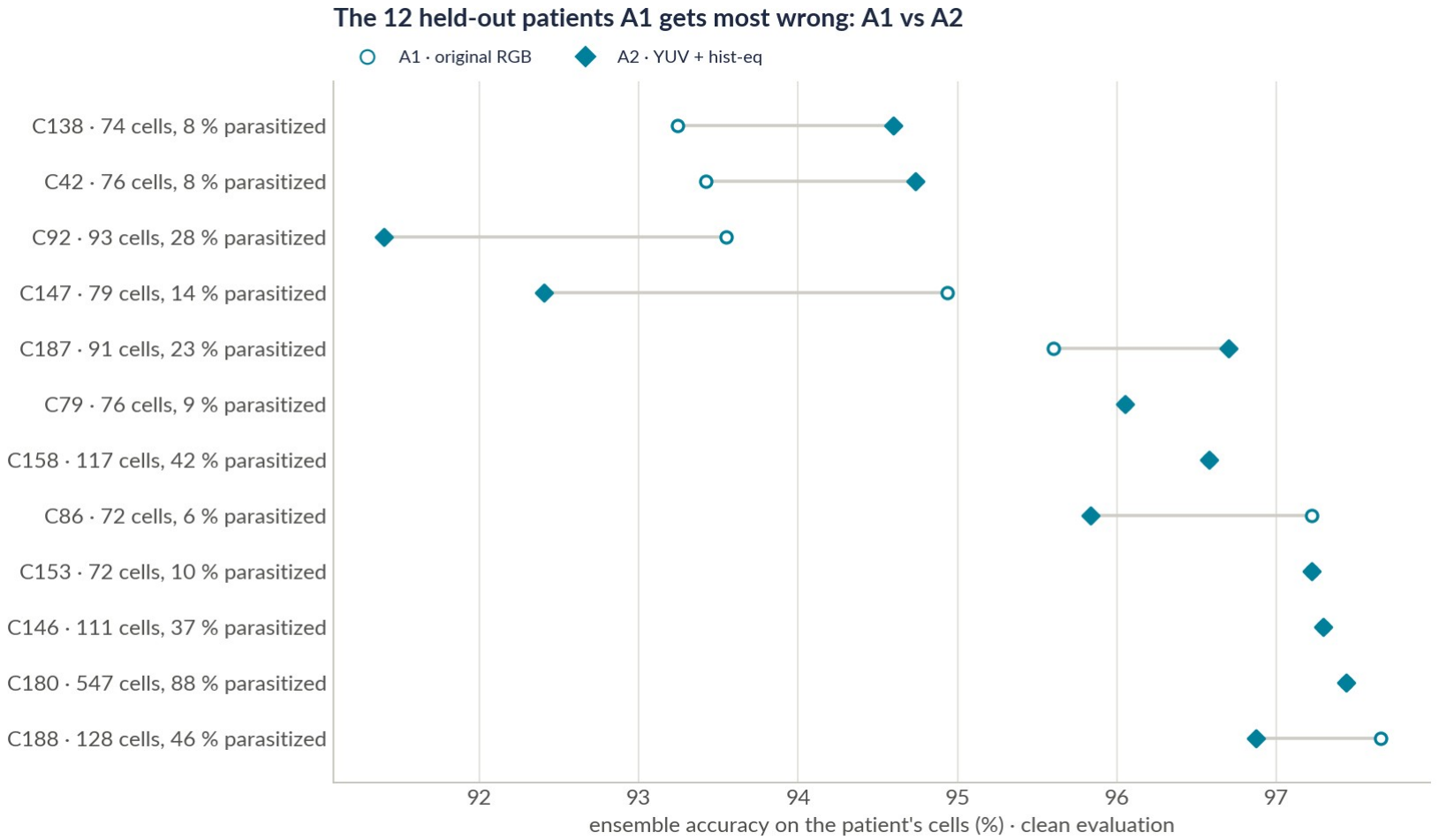
held-out patients account for half of A1's 86 errors

14 / 40

unseen patients with every cell classified correctly

30 of 31

false alarms fall on infected patients' smears (1.57 % vs 0.12 % of uninfected cells)



SCREENING VIEW · NO CLINICAL CLAIM

Recall first: a missed infection costs more than a false alarm

Experiment	Threshold (chosen on CV folds)	Hold-out recall at 0.5 → at threshold	Specificity at threshold	ECE
A0 · image split	0.192	97.18 → 98.72 %	95.22 %	0.053
A1 · patient split	0.189	98.05 → 99.22 %	93.93 %	0.065
A2 · patient split	0.143	98.22 → 99.22 %	91.74 %	0.060
A3 · image split	0.159	97.14 → 98.75 %	94.14 %	0.052



Tune for recall

The decision threshold is chosen on the 10 CV folds (best F2), never on the hold-out set, then applied once.



Calibration

Label smoothing (0.1) caps confidence near 0.95, so probabilities read as under-confident (ECE ≈ 0.05).

Combined per patient: 26 of 28 infected, 12 of 12 healthy

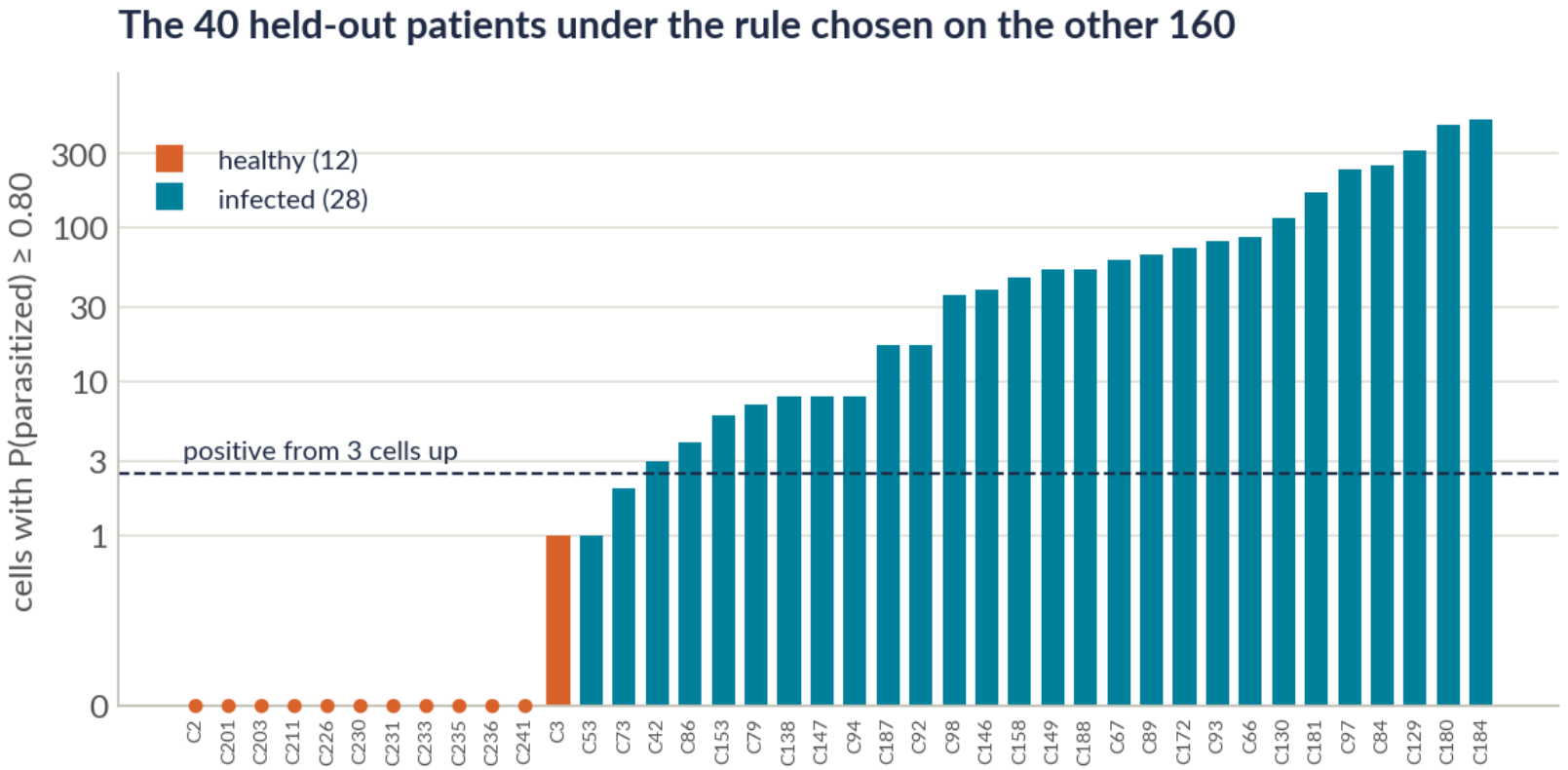
26 / 28

infected held-out patients found (92.9 %; 95 % CI 76.5 to 99.1)

12 / 12

healthy held-out patients cleared (95 % CI 73.5 to 100.0 %)

Rule chosen on the 160 CV patients only: positive if at least 3 cells have P ≥ 0.80. Exploratory: NIH curated the crops, so this is not diagnostic accuracy on a smear.



HONEST LIMITS

What these results do not show

Curated cells, not smears

Our patient-level rule is exploratory: NIH kept about 68 uninfected crops per patient and curated the parasitized ones, so it is not diagnostic accuracy on a smear.

One hospital

A patient split removes patient leakage, not site or stain-protocol effects; there is no external test set.

One expert's labels

Fuhad et al. set aside 5.1 % of cells as mislabelled or suspicious, and our false alarms concentrate on infected smears. We keep the original labels, as the paper did.

One split, one run

A different draw of 40 hold-out patients, or a second training seed, would move the numbers; the CIs show by how much.

CONCLUSION

What we take away

1

Reproducible

Rebuilt from the authors' code, the ensemble lands within 1 pp of the paper on every metric, trained 25× faster.

2

98 % survives unseen patients

98.45 % on 40 held-out patients. Leakage is real but small: 0.85 pp of MCC on identical cells. H1 is not supported.

3

YUV doesn't help

Stain normalisation changes MCC on unseen patients by −0.26 pp, within ±1 pp of zero. H2 is rejected.

Next

- An external test set from another lab (MP-IDB)
- Blind triage of our errors; compare with an independent relabelling
- Patient-level rules on whole smears
- Distil the ensemble into one light model

Thank you · questions welcome · github.com/amathziah/malaria-detection

REFERENCES

Sources

[1] Z. Liang et al., “CNN-based image analysis for malaria diagnosis,” IEEE BIBM, 2016.

[2] S. Rajaraman et al., “Pre-trained CNNs as feature extractors toward improved malaria parasite detection,” PeerJ 6:e4568, 2018.

[3] S. Rajaraman, S. Jaeger, S. K. Antani, “Performance evaluation of deep neural ensembles toward malaria parasite detection,” PeerJ 7:e6977, 2019.

[4] M. Umer et al., “A novel stacked CNN for malarial parasite detection in thin blood smear images,” IEEE Access 8, 2020.

[5] K. M. F. Fuhad et al., “Deep learning based automatic malaria parasite detection from blood smear and its smartphone based application,” Diagnostics 10(5):329, 2020.

[6] O. S. Zhao et al., “Convolutional neural networks to automate the screening of malaria in low-resource countries,” PeerJ 8:e9674, 2020.

[7] M. Masud et al., “Leveraging deep learning techniques for malaria parasite detection using mobile application,” WCMC 2020:8895429.

[8] G. Marques, A. Ferreras, I. de la Torre-Díez, “An ensemble-based approach for automated medical diagnosis of malaria using EfficientNet,” MTAP 81, 2022 (base paper).

[9] G. Madhu et al., “Intelligent diagnostic model for malaria parasite detection … capsule neural networks,” Sci. Rep. 13, 2023.

[10] K. Hemachandran et al., “Performance analysis of deep learning algorithms in diagnosis of malaria disease,” Diagnostics 13(3):534, 2023.

[11] M. Mujahid et al., “Efficient deep learning-based approach for malaria detection using red blood cell smears,” Sci. Rep. 14, 2024.

[12] C. Nadeau, Y. Bengio, “Inference for the generalization error,” Machine Learning 52, 2003 · [13] S. Holm, “A simple sequentially rejective multiple test procedure,” Scand. J. Statist. 6, 1979.

[14] D. Lakens, “Equivalence tests: A practical primer for t tests, correlations, and meta-analyses,” Soc. Psychol. Personal. Sci. 8(4), 2017 · [15] C. G. Northcutt, L. Jiang, I. L. Chuang, “Confident learning: Estimating uncertainty in dataset labels,” JAIR 70, 2021.

[16] D. Tellez et al., “Quantifying the effects of data augmentation and stain color normalization in convolutional neural networks for computational pathology,” Med. Image Anal. 58, 2019.